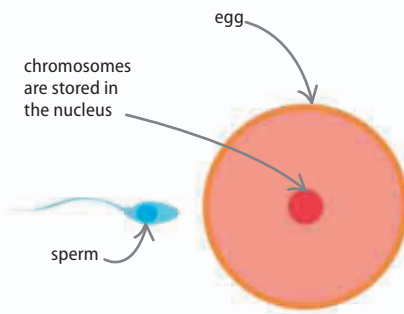


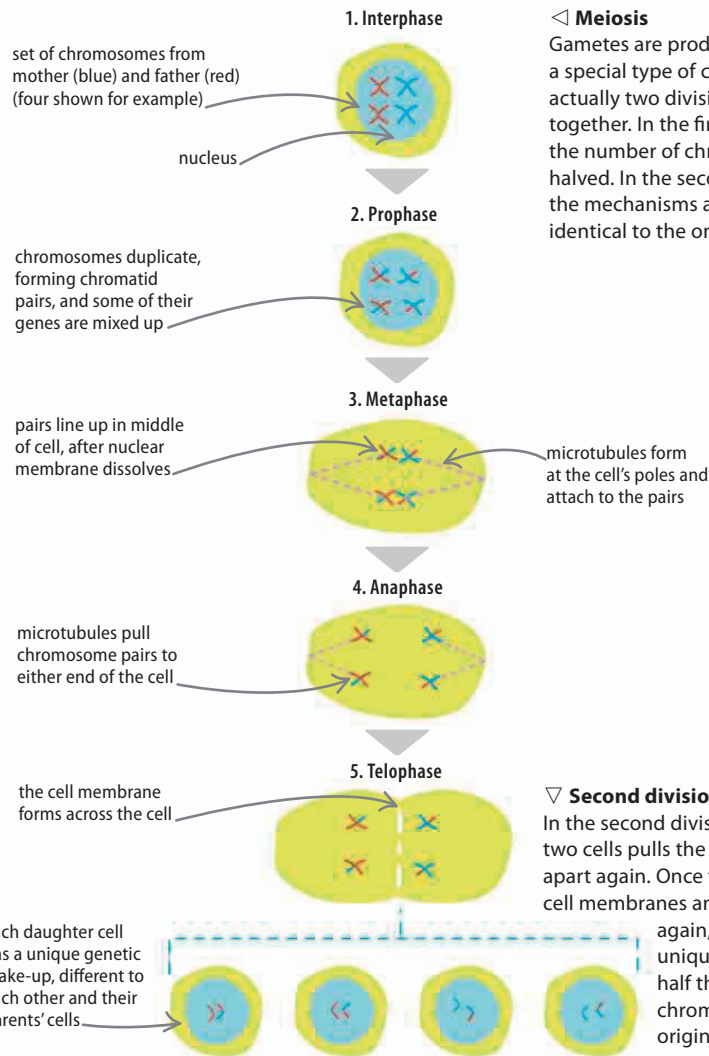
Sexual reproduction

Sexual reproduction happens when two parents mix up their genes in order to produce offspring with their own unique genetic make-ups. Sexual reproduction requires each parent to produce gametes, or sex cells. While ordinary cells contain two full sets of genes—one from each parent—gametes have just a single set. In a process called fertilization, two gametes—one from each parent—fuse to form a zygote, the first cell of a new individual.



△ Sperm and egg

Male gametes are called sperm, and female ones are called eggs or ova (singular: ovum). Both contain half the usual number of chromosomes. A sperm's purpose is to deliver genes to the egg and it contains nothing else. By contrast, an egg cell needs to be huge to contain the nutrients required to grow a new individual after fertilization.



◁ Meiosis

Gametes are produced using a special type of cell division—actually two divisions occurring together. In the first division, the number of chromosomes is halved. In the second division, the mechanisms are almost identical to the ones in mitosis.

▽ Second division

In the second division, each of the two cells pulls the chromosomes apart again. Once this is done, the cell membranes and nuclei form again, leaving four unique cells, with half the number of chromosomes of the original parent cells.

Animal development

After fertilization, the new individual (embryo) needs to develop and grow until it is ready to feed and live independently. The ways that animals produce their young depends on their habitat and biology.

▷ Development strategies

Small creatures, which are under constant threat of predators, produce lots of young quickly. Larger and better protected animals invest in protecting fewer young instead.

Method	Explanation	Example
ovuliparity	eggs are fertilized after being released by the female	fish, toads
oviparity	eggs are fertilized before release and often protected in a nest	birds
ovoviviparity	fertilized eggs retained in body until after hatching	seahorses
aplacental viviparity	young grow inside mother, feeding on eggs or siblings	some sharks
placental viviparity	young sustained by mother through placenta until birth	mammals